Breaking Records:
Are Record Earnings a Reference Point for Investors?

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Abstract

A firm's best-ever earnings are a natural reference point: they are salient, easy to compute and often reported as "record profits". We test whether investors, analysts and managers treat them as one, using 95,270 quarterly reports of Chinese A-share firms from 2005 to 2020. A standard regression discontinuity at the record seems to find anchoring. Returns do not jump at the record, but the drift after the report steepens once earnings pass it. We show that this pattern is not caused by the record. Trailing earnings change by exactly the year-on-year surprise, so the distance to the record equals the surprise minus the gap to the record before the report. For 72% of records the gap is zero, and setting a record is the same event as beating last year's quarter. The same design also finds "effects" of the record on firm size, which the record cannot cause. We then use the fact that the threshold moves with the gap: two firms with the same surprise land on different sides of their record only because they started at different distances from it. A new record then changes neither returns nor trading volume, analyst forecasts or future earnings. We can rule out a record premium larger than the price response to a one-standard-deviation surprise. Firms do not bunch just above their record, although they bunch strongly just above zero earnings. Where the record seems to matter, it marks a state in which good news persists and bad news does not, and prices learn this only slowly.

Keywords: reference points, anchoring, record earnings, regression discontinuity, earnings management, China. **JEL:** G12, G14, G41, M41.

1 Introduction

"Record profit" is one of the most common headlines in financial news. Records are what reference-dependent preferences are about. People evaluate outcomes relative to a reference point (Kahneman and Tversky, 1979; Kőszegi and Rabin, 2006), treat goals and personal bests as reference points (Heath et al., 1999; Pope and Simonsohn, 2011; Allen et al., 2017), and pay attention to what is salient (Bordalo et al., 2012). In asset markets, investors anchor on salient past prices such as the 52-week high and the all-time high (George and Hwang, 2004; Huddart et al., 2009; Li and Yu, 2012; Baker et al., 2012). Managers, in turn, care about earnings thresholds such as zero earnings and last year's earnings (Burgstahler and Dichev, 1997; Degeorge et al., 1999; Graham et al., 2005), and the market rewards strings of earnings increases (Barth et al., 1999; Myers et al., 2007). A firm's best-ever earnings combine all of these: a salient past level that new earnings either pass or fail to pass. Jang and Lee (2021) find that the drift after earnings announcements is stronger when earnings are close to their record and read this as anchoring.

Testing whether a record *causes* anything is harder than it looks. Firms that set records are different. They are larger, have better momentum and receive better news (Table 2). A regression

discontinuity (RD) at the record should take care of this, by comparing reports that just set a record with reports that just miss one. The difficulty is that the record is reached through the news itself, and this has a simple consequence. Trailing twelve-month (TTM) earnings per share change in each report by exactly this quarter's EPS minus the EPS of the same quarter a year earlier: the surprise relative to a seasonal random walk. Measured in the same units, the distance to the record after the report is

$$r = SUE - gap, \tag{1}$$

where $gap \geq 0$ is how far the firm's TTM earnings were below its record before the report. For example, on 27 April 2018 Wuliangye, a large liquor maker, reported first-quarter EPS of 1.30 yuan, against 0.95 a year earlier: a surprise of 3.7 standard deviations. Its TTM EPS had been 2.8 standard deviations below the record it set in 2013, before the anti-graft campaign hit liquor sales. The surprise exceeded the gap, and the firm set a new record by $r = 3.7 - 2.8 = 0.9$ (Figure 1a).

The identity has two consequences. First, 72% of all records are set by firms whose previous report was already a record ($gap = 0$). For them $r = SUE$, and setting a record is the same event as beating last year's quarter. Second, for firms below their record, the cutoff moves with the gap, which is known before the report.

The first consequence explains a misleading result. A standard RD at the record (the design of our proof of concept) finds no jump in returns, but finds that the drift over the next three months rises faster with r above the record than below it (a "kink" of 3.09% per standard deviation, $t = 2.51$). This looks like anchoring: investors seem to underreact to news that pushes earnings into new territory. We show it is not a record effect. The kink comes entirely from firms already at their record, for which it is a property of the response to ordinary good and bad news. Among firms below their record it is 0.34 ($t = 0.19$). And the same design finds equally strong kinks and jumps in variables fixed before the report, such as firm size ($t = 5.56$) and the previous quarter's surprise ($t = -3.89$), which the record cannot cause.

The second consequence gives a clean design. Consider two firms that both report earnings one standard deviation above last year's quarter (Figure 1b). Firm A was 0.8 standard deviations below its record, so it now sets a new record; firm B was 1.2 below, so it falls just short. Their news is the same; only their starting points differ, and those were fixed before the report. We give every narrow band of surprises its own intercept and compare firms just above and just below their record within each band. In this *moving-cutoff* design, a new record changes nothing we can measure. The announcement-month return rises by an insignificant 0.34% ($t = 0.71$), and the drift, trading volume, analyst forecast revisions, the next quarter's surprise and earnings growth over the next year do not jump either. We can rule out a record premium larger than 0.97%, about the price response to a one-standard-deviation earnings surprise. Characteristics fixed before the report are balanced across the record.

Managers do not target records either. The density of reports is smooth at the record (excess mass 3.6%, $t = 1.11$). The same test finds 153% excess mass just above zero TTM earnings and 1,394% just above zero annual earnings, the well-known loss avoidance induced by China's delisting rules (Chen and Yuan, 2004; Haw et al., 2005).

What, then, is special about firms at their record? Their news behaves differently. When they report good news, it persists into the next quarter much more strongly than for firms far below their record; when they report bad news, it does not persist at all. Prices at the announcement treat both groups alike, and the drift then follows the persistence: bad news at the peak is reversed over the next three months, while good news keeps drifting. This is the pattern behind the naive

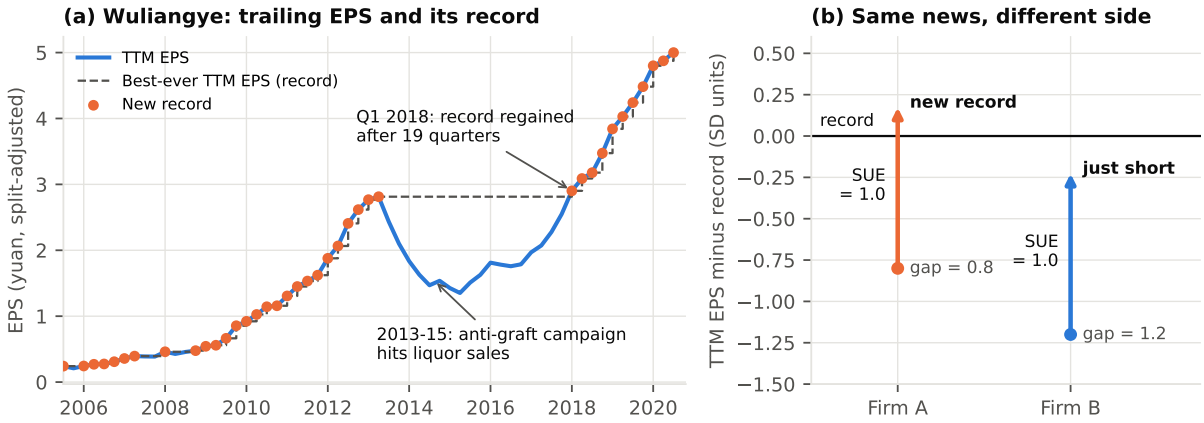


Figure 1: What it takes to set a record. (a) Wuliangye's trailing twelve-month (TTM) EPS, its record (best-ever TTM EPS, dashed) and the reports that set a new record (dots). The firm regained its 2013 record in its first-quarter report of 2018, 19 quarters later. (b) Two firms with the same surprise ($SUE = 1$) but different gaps to their record before the report. Firm A sets a record; firm B falls just short. Our main design compares such firms.

kink. It suggests a different reading of evidence based on proximity to the record. Investors do not seem to anchor on the record; if anything, they ignore what being at the record says about the persistence of news.

The paper makes three contributions. First, it gives clean evidence on a natural candidate reference point and finds that it does not matter for prices, trading, analysts or reporting, with bounds that exclude economically meaningful effects. This contrasts with the evidence on price anchors (George and Hwang, 2004; Li and Yu, 2012) and on earnings thresholds such as zero and last year's earnings (Burgstahler and Dichev, 1997; Degeorge et al., 1999), and qualifies the anchoring interpretation of Jang and Lee (2021). Second, it shows a general pitfall and a general fix. Any threshold defined by a running maximum, such as a record, a 52-week high or a fund's high-water mark, is crossed through the news itself, and most crossings are made by units that were already at the maximum. A regression discontinuity in the distance to the maximum then mixes the threshold with the shape of the response to news. The fix is to use the distance before the news, which is fixed in advance, as the source of variation (Papay et al., 2011; Reardon and Robinson, 2012; Wong et al., 2013). Third, it adds to the evidence on earnings management and investor behaviour in China, a market dominated by retail investors who trade on attention (Seasholes and Wu, 2007; Carpenter et al., 2021; Jones et al., 2025). Even there, the record does not attract trading or move prices.

2 Hypotheses

If investors use the record as a reference point, crossing it should change how they react to the same news. The literature suggests four testable versions.

- **H1 (record premium).** A record is a salient piece of good news and a goal reached (Heath et al., 1999; Bordalo et al., 2012). Prices jump when earnings just pass the record. If the premium is an overreaction, it reverses later.
- **H2 (anchoring).** Investors anchored on the record are reluctant to believe that earnings will stay above it, so they underreact to news that pushes earnings past it. Prices then catch up later, as in the 52-week-high effect (George and Hwang, 2004; Jang and Lee, 2021). Returns after the report jump at the record, or rise faster with the margin by which the record is

beaten.

- **H3 (attention).** Records attract attention, especially from retail investors (Barber and Odean, 2008; Seasholes and Wu, 2007), and trading volume rises around salient price levels (Huddart et al., 2009). Turnover jumps at the record.
- **H4 (targeting).** If records matter to investors, managers have an incentive to report just above them, as they do for zero earnings and last year's earnings (Burgstahler and Dichev, 1997; Degeorge et al., 1999; Myers et al., 2007). Reports bunch just above the record.

We also test whether analysts revise their forecasts differently after a record, and whether records predict future earnings. Under the null hypothesis, the record is irrelevant once we know the news.

3 Data

Sample. We use monthly data on all stocks traded on the Shanghai and Shenzhen exchanges from Thomson Reuters Datastream and Worldscope, including inactive stocks to avoid survival bias (Shumway, 1997). The data give prices, total return indices, trading volume, shares outstanding, book values, the I/B/E/S consensus forecast, and standalone quarterly EPS with each report's publication date. Our sample contains reports published from 2005 to 2020. Table 1 shows how the 95,270 reports of 2,817 firms are selected.

Records. Trailing twelve-month EPS (TTM) is the sum of the last four standalone quarters, adjusted for stock splits and bonus shares so that it is comparable within a firm over time. The *record* is the highest TTM EPS of all earlier reports; we require at least eight earlier values, so that a record means something, and a positive record. We measure the news in each report against a seasonal random walk (SRW): expected EPS equals EPS of the same quarter a year earlier (Foster, 1977; Bernard and Thomas, 1990). The scale σ is the standard deviation of the firm's last eight SRW errors, and all distances are in units of σ:

$$SUE = \frac{EPS_q - EPS_{q-4}}{\sigma}, \qquad r = \frac{TTM_{\text{new}} - \text{record}}{\sigma}, \qquad gap = \frac{\text{record} - TTM_{\text{before}}}{\sigma} \geq 0.$$

Because $TTM_{\text{new}} - TTM_{\text{before}} = EPS_q - EPS_{q-4}$, identity (1) holds exactly. A report sets a new record when $r > 0$, that is, when the surprise exceeds the gap. We drop 451 exact ties with the record. In the Wuliangye example, $\sigma = 0.094$ yuan, TTM EPS rose from 2.55 to 2.90 yuan and the old record was 2.81 yuan.

Outcomes. Returns are monthly and size-adjusted: the stock's return minus the average return of its size decile. Month 0 is the calendar month of the report, so it contains the reaction to it; we also use the drift over months +1 to +3 and +1 to +6. Abnormal turnover is log monthly turnover minus the firm's average over the previous year. The analyst revision is the change in the I/B/E/S 12-month-forward consensus EPS from the start of the report month to the start of the next month, in basis points of the price. We drop April reports here, because the consensus switches to a new fiscal year in May. Future earnings are the next quarter's SUE and the change in TTM EPS over the next four quarters, in units of σ.

Descriptive statistics. Records are common. 20% of reports set one, and 80% of firms set at least one. Most records extend a run: the previous report was already a record (*extended* records,

Table 1: Sample construction

Step	Reports
Reports with an SRW surprise, 2005–2020	106,679
Four consecutive quarters, now and one quarter earlier	104,661
At least eight earlier TTM values (record defined)	95,807
Positive record	95,721
Not an exact tie with the record (main sample)	95,270
New records ($r > 0$)	19,316
of which the previous report was the record ($gap = 0$)	13,868
Firms	2,817

Reports of A-share firms with a seasonal-random-walk (SRW) surprise. TTM: trailing twelve-month EPS.

Table 2: Descriptive statistics by record status

		New record		No record	
	All	Extended	Regained	Near miss	Far below
Reports	95,270	13,868	5,448	13,179	62,775
Share of reports (%)	100.0	14.6	5.7	13.8	65.9
Surprise, SUE	0.10	1.53	2.21	0.22	−0.42
Gap to record before report (median)	2.16	0.00	0.63	0.46	3.68
Quarters since record (median)	8	1	6	4	11
Log market value	15.87	16.51	16.15	15.85	15.71
Log book-to-market	−1.42	−1.73	−1.57	−1.55	−1.31
Momentum, months −12 to −2 (%)	15.6	40.2	33.6	20.5	7.6
Price / 52-week high	0.80	0.85	0.84	0.81	0.79
Analyst coverage (%)	50	73	57	48	45
Annual report (%)	24	21	33	24	24
Return, announcement month (%)	0.01	1.21	2.33	−0.13	−0.44
Return, months +1 to +3 (%)	−0.59	0.90	1.07	−0.68	−1.06
Abnormal turnover (%)	7.8	4.9	10.3	7.4	8.4
Next-quarter SUE	0.11	0.79	0.44	−0.01	−0.05

Means (medians where stated). Extended: new record by a firm at its record. Regained: new record by a firm below it. Near miss: $-1 \le r < 0$. Far below: $r < -1$. Returns are size-adjusted.

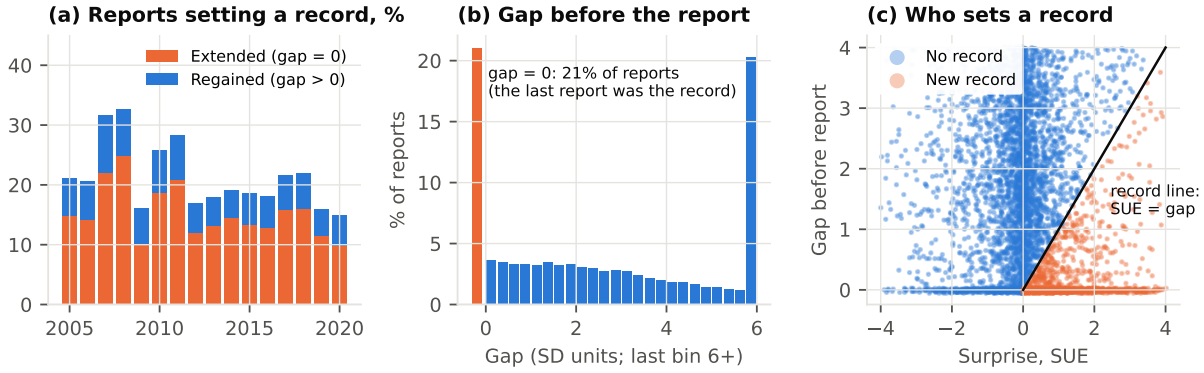


Figure 2: Records in the data. (a) Share of reports that set a new record, by year. (b) Gap to the record before the report, in units of σ; the separate bar is the share of reports that start at their record. (c) A random sample of 5,000 reports by surprise and gap. Records lie below the line $SUE = gap$; points at $gap = 0$ are jittered.

Figure 2a). Only 5.7% of reports *regain* a record after falling below it, typically 6 quarters after it was set. Figure 2b shows the distribution of the gap: 21% of reports start at their record, and the rest are spread out with a median gap of 3.1. Panel (c) shows who sets a record in the (surprise, gap) plane. Firms at the bottom line ($gap = 0$) set a record whenever $SUE > 0$; firms above it need a surprise larger than their gap. Table 2 shows that record setters are larger and more often covered by analysts, have much stronger momentum (40% against 8% for firms far below their record) and earn higher returns. Simple comparisons therefore mix the record with good news and good firms.

4 Methodology

The naive RD. The obvious design is a local-linear RD in the distance to the record:

$$y_i = \alpha + \tau D_i + \beta r_i + \kappa D_i r_i + \gamma_+ SUE_i^+ + \gamma_- SUE_i^- + \varepsilon_i, \qquad |r_i| \le h, \tag{2}$$

where $D_i = 1[r_i > 0]$ marks a record, the weights fall linearly to zero at the bandwidth h (a triangular kernel), and SUE^+, SUE^- control for good and bad news with separate slopes. The jump τ tests H1. The kink κ, the change in slope at the record, tests the margin version of H2: does the return rise faster with r once earnings pass the record?

Equation (2) has two problems. For firms at their record, $r = SUE$, so $D_i r_i = SUE_i^+$. Their "record kink" is any change in the slope of the response to news at $SUE = 0$, and returns are known to respond to news in a nonlinear way (Freeman and Tse, 1992). And kinks are fragile: any curvature in how an outcome varies with r shows up as a kink, because the two linear fits differ in slope even without a break (Card et al., 2015). A balance test on variables fixed before the report reveals both problems.

Main design: the moving cutoff. We restrict the sample to firms below their record ($gap > 0$) and estimate

$$y_i = \alpha_{b(i)} + \tau D_i + \beta r_i + \kappa D_i r_i + \varepsilon_i, \qquad |r_i| \le h, \tag{3}$$

where $b(i)$ is the surprise bin of report i (bins of 0.25σ), so each bin has its own intercept. Within a bin the surprise is almost constant, and by identity (1) the distance to the record varies only through the gap. The jump τ therefore compares firms with the same news that end up on different sides of their record because they started at different distances. The identifying assumption is that, given the surprise, outcomes would change smoothly with the gap at the point where the gap equals the surprise. This is plausible because the gap is fixed before the report and the cutoff it implies differs from firm to firm. It would fail if firms could manage earnings to just pass the record, which the bunching test checks. In the language of the multi-score RD literature, records are a two-score design in the surprise and the gap with the boundary $SUE = gap$, and we estimate the effect along this boundary (Papay et al., 2011; Reardon and Robinson, 2012; Wong et al., 2013).

The jump is our main estimate. We use $h = 1$ and report $h = 2$ and a local-quadratic fit at $h = 2$, which corresponds to the robust bias correction of Calonico et al. (2014); we avoid higher-order polynomials (Gelman and Imbens, 2019). We do not interpret kinks, for the reason given above. Standard errors are clustered by report month and by firm (Petersen, 2009; Cameron et al., 2011).

Firms at their record: a difference in discontinuities. For firms at their record, the record cannot be separated from beating last year's quarter within the firm group. We therefore compare them with firms far below their record ($gap > h$), for which no surprise within the window sets a record, at the same cutoff $SUE = 0$ (Grembi et al., 2016):

$$y_i = \alpha_m + \sum_{g \in \{0,1\}} 1[G_i = g]\left(a_g + \tau_g D_i + \beta_g SUE_i + \kappa_g D_i SUE_i\right) + \varepsilon_i, \qquad |SUE_i| \le h, \tag{4}$$

where now $D_i = 1[SUE_i > 0]$, $G_i = 1$ for firms at their record, and α_m are report-month fixed effects. The differences $\tau_1 - \tau_0$ and $\kappa_1 - \kappa_0$ measure how the response to beating last year's quarter changes when it also sets a record. The comparison is between two kinds of firms, so we read it together with the evidence on their fundamentals.

Bunching. To test H4 we count reports in bins of 0.05σ and fit a Poisson regression of the counts on a polynomial in the running variable on each side of the threshold. The excess mass just above the threshold is $e^{\hat{\delta}} - 1$, where $\hat{\delta}$ is the jump in the log density (McCrary, 2008; Cattaneo et al., 2020; Kleven, 2016).

Table 3: The naive regression discontinuity at the record

A. Size-adjusted returns (%)	Announcement month		Months +1 to +3		Months +1 to +3	
	All reports				Gap > 0 only	
	$h = 1$	$h = 2$	$h = 1$	$h = 2$	$h = 1$	$h = 2$
Jump at the record	−0.13	0.02	0.34	0.19	1.06	0.74
	(−0.44)	(0.08)	(0.72)	(0.54)	(1.47)	(1.38)
Kink: change in slope	1.08	0.76***	3.09**	1.45***	0.34	0.41
	(1.45)	(2.65)	(2.51)	(2.68)	(0.19)	(0.50)
Reports	22,753	39,200	22,182	38,198	12,182	23,543

B. Placebo outcomes fixed before the report (all reports, $h = 1$)	Log size	Momentum (%)	Price/52w high	Previous SUE	Pre-report return (%)
Jump at the record	0.12***	2.26	0.01**	0.23***	0.30
	(3.40)	(1.28)	(2.26)	(3.53)	(0.69)
Kink: change in slope	0.74***	−5.63	0.02	−0.48***	0.83
	(5.56)	(−0.80)	(1.46)	(−3.89)	(0.75)

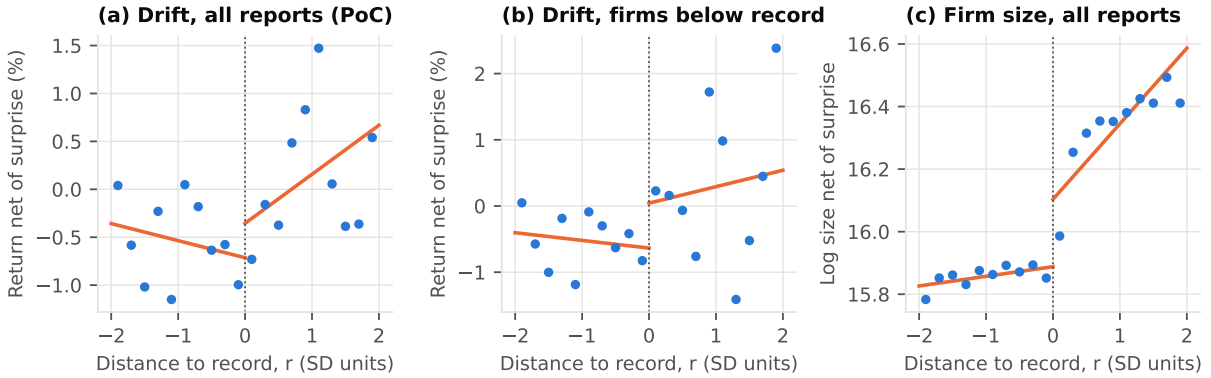


Figure 3: The naive design. Bin means of the outcome net of good and bad surprises, by distance to the record, with linear fits on each side. (a) Drift, all reports. (b) Drift, firms below their record before the report. (c) Log market value at the start of the report month, a placebo outcome.

Equation (2): local-linear RD with triangular kernel and bandwidth h (in units of σ), controlling for SUE^{+} and SUE^{-}. Panel B uses the same specification with variables fixed before the report as outcomes. t-statistics in parentheses, clustered by report month and firm. *, **, ***: 10%, 5%, 1%.

5 Results

5.1 The naive RD finds a kink, but also in firm size

Table 3 reproduces the naive design. Returns do not jump at the record in the report month (−0.13%, $t = -0.44$) or later. The drift over months +1 to +3, however, rises faster with r above the record than below it, by 3.09% per unit of σ ($t = 2.51$) at $h = 1$ and by 1.45% ($t = 2.68$) at $h = 2$. Figure 3a shows the pattern: below the record the drift net of the surprise is flat or falling; above it, it rises. Read naively, investors underreact to news that pushes earnings past the record.

Two facts show that this reading is wrong. First, the kink disappears among firms below their record, where the record and beating last year differ (0.34, $t = 0.19$; Figure 3b). Second, the same design applied to variables that are fixed before the report (Panel B) finds a jump and a kink in firm size ($t = 5.56$ for the kink; Figure 3c) and in the previous quarter's surprise ($t = -3.89$). The record cannot change how large a firm was before the report. The design compares firms that differ in more than the record.

Table 4: Main result: firms below their record, same surprise, different side of the record

	Return, month 0 (%)	Return, +1 to +3 (%)	Return, +1 to +6 (%)	Abnormal turnover (%)	Analyst revision (bp)	Next-quarter SUE	EPS growth, 4 quarters
Jump at the record, $h = 1$	0.34	0.87	−1.74	−0.73	−1.92	−0.08	−0.31
	(0.71)	(1.20)	(−1.64)	(−0.22)	(−0.91)	(−1.28)	(−1.61)
Jump at the record, $h = 2$	0.29	0.60	−1.21	−2.26	−0.53	−0.10*	−0.36**
	(0.84)	(1.06)	(−1.51)	(−0.92)	(−0.33)	(−1.80)	(−2.04)
Local quadratic, $h = 2$	0.41	0.89	−2.01*	−0.47	−1.62	−0.08	−0.33
	(0.81)	(1.15)	(−1.78)	(−0.14)	(−0.68)	(−1.14)	(−1.56)
Mean just below the record	0.83	−0.36	0.24	9.48	6.21	0.13	0.17
Reports ($h = 1$)	9,859	9,573	9,276	9,282	3,273	9,951	9,149

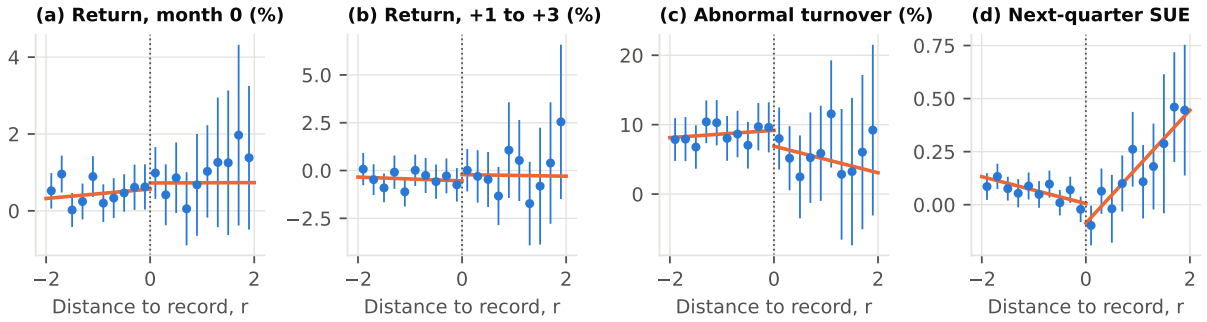


Figure 4: Outcomes around the record, firms below their record. Bin means net of surprise-bin effects with 95% confidence intervals, and local-linear fits on each side.

Equation (3): sample of reports with $gap > 0$; every surprise bin of 0.25σ has its own intercept. Local-linear (or quadratic) RD with triangular kernel and bandwidth h. Returns and turnover in %, analyst revisions in basis points of the price, EPS growth in units of σ. t-statistics in parentheses, clustered by report month and firm.

5.2 Main result: the same news, a different side of the record

Table 4 reports the moving-cutoff design (3) for the 75,240 reports of firms below their record. None of the seven outcomes jumps at the record. The announcement-month return is 0.34% higher ($t = 0.71$), the drift over months +1 to +3 is 0.87% higher ($t = 1.20$), and the drift over six months is, if anything, lower (−1.74%, $t = -1.64$). Trading volume does not rise (−0.73%, $t = -0.22$), so a record does not attract attention (H3). Analysts do not revise their forecasts differently (−1.92 bp, $t = -0.91$). Nor does a record predict future earnings: the next quarter's surprise and the next year's earnings growth are, if anything, slightly lower above the record. A record therefore carries no information about the future beyond the surprise, and the market correctly ignores it. Figure 4 shows the raw data: the bin means run smoothly through the record.[1]

How large an effect can we rule out? At $h = 2$ the 95% confidence interval for the announcement-month premium is [−0.38, 0.97]%. Across all reports, a one-standard-deviation surprise moves the announcement-month return by 0.97%, so a record premium larger than the reaction to a 1.0-standard-deviation surprise is ruled out. For the three-month drift the interval is [−0.51, 1.70]%.

Table 5 supports the design. Characteristics fixed before the report do not jump at the record at $h = 1$; the largest t-statistic is 1.09. At $h = 2$ the previous quarter's surprise is slightly lower just above the record ($t = -3.12$), but this disappears with a local-quadratic fit, which suggests curvature rather than a break; controlling for it leaves the results unchanged (Appendix Table A1). Figure 5 shows that the estimates do not depend on the bandwidth.

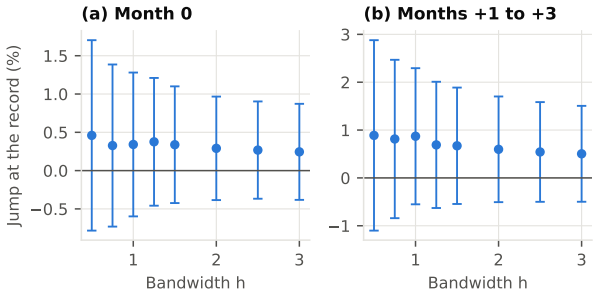
[1]Within a surprise bin, a larger r means a smaller gap, so the slopes in Figure 4 show how outcomes vary with the gap. These slopes differ across the record also for variables fixed before the report, such as firm size, so we do not interpret kinks in this design either.

Table 5: Balance: variables fixed before the report

	$h = 1$		$h = 2$	
	Jump	t	Jump	t
Log size	0.056	(1.09)	0.066	(1.64)
Log B/M	−0.029	(−0.76)	−0.030	(−1.00)
Momentum (%)	1.806	(0.72)	2.473	(1.04)
Price/52w high	−0.001	(−0.17)	−0.006	(−0.95)
Return, −2 to −1 (%)	−0.413	(−0.61)	−0.083	(−0.16)
Previous SUE	−0.033	(−0.75)	−0.109***	(−3.12)
Analyst coverage	−0.002	(−0.09)	0.008	(0.50)
Annual report	0.003	(0.17)	−0.005	(−0.31)

Each row re-estimates Equation (3) with a characteristic measured before the report as the outcome.

Figure 5: Bandwidth choice



Jump at the record (Equation (3)) and 95% confidence intervals for bandwidths from 0.5 to 3.

5.3 Firms at their record: different news, the same price response

Firms whose previous report was a record set a new one whenever they beat last year's quarter. Table 6 compares their response at $SUE = 0$ with that of firms far below their record (Equation (4)), and Figure 6 plots both. Three results stand out.

First, beating last year's quarter earns no extra premium when it also sets a record. The extra jump is −0.66% ($t = -1.58$) in the report month and −0.06% ($t = -0.08$) later. This is the same conclusion as in the main design, from a different comparison.

Second, the news itself differs. For firms far below their record, a one-σ surprise predicts a next-quarter surprise of 0.62 when the news is bad and 0.39 when it is good. For firms at their record, bad news does not persist at all (0.03, $t = 0.17$), while good news persists strongly (0.88, $t = 10.13$; Figure 6c). A weak quarter at the peak is usually a blip; a strong quarter at the peak is usually the start of more.

Third, prices learn this slowly. In the report month, the response of firms at their record has the same shape as that of other firms (Figure 6a; the extra kink is −0.25, $t = -0.21$). Over the next three months, the drift of firms at their record follows the persistence of their news. Bad news tends to be reversed (−1.92% per σ, $t = -1.59$), while good news keeps drifting (3.99% per σ, $t = 3.58$). The resulting extra kink, 5.05% ($t = 2.92$), is the naive kink of Table 3. It survives when the comparison firms also had good news in the previous quarter (column 4). Controlling for the realised next-quarter surprise explains about 24% of it (column 6), a lower bound because only part of the next reports fall inside the three-month window.

This is not the anchoring of H2, under which investors would treat the record as special and underreact to news that passes it. The evidence points the other way: at the announcement, investors treat news at the record like any other news, even though it is more (or less) persistent, as in the classic drift of Bernard and Thomas (1989, 1990). The contrast matters for proximity-based evidence such as Jang and Lee (2021). Drift that is stronger near the record can arise because news near the record is more persistent, not because the record is an anchor.

5.4 Do firms manage earnings to set records?

If records mattered to investors, managers would want to report just above them (H4). Table 7 and Figure 7 show that they do not. The density of reports is smooth at the record (3.6%, $t = 1.11$), also among firms far below their record, where the threshold lies away from the cluster of small surprises around zero (5.7%, $t = 0.45$). The test is not powerless. It finds 153% excess mass just above zero TTM earnings ($t = 11.04$) and 1,394% just above zero annual earnings ($t = 26.05$). In China, two consecutive annual losses lead to special treatment and, eventually,

Table 6: Firms at their record versus firms far below it, at $SUE = 0$

	(1) Month 0	(2) +1 to +3	(3) +1 to +3	(4) +1 to +3	(5) Next SUE	(6) +1 to +3
Slope below SUE = 0, far-below firms	0.72**	1.57**	1.07***	0.97	0.62***	1.00
	(1.97)	(2.32)	(2.85)	(0.95)	(5.39)	(1.48)
Extra slope, at-record firms	1.19	−3.49**	−1.57**	−2.87*	−0.59***	−2.94**
	(1.33)	(−2.42)	(−2.48)	(−1.75)	(−3.94)	(−2.05)
Jump at SUE = 0, far-below firms	0.36*	0.16	0.29	0.35	0.05*	0.10
	(1.65)	(0.45)	(1.01)	(0.76)	(1.96)	(0.28)
Extra jump, at-record firms	−0.66	−0.06	−0.05	−0.30	−0.07	0.02
	(−1.58)	(−0.08)	(−0.08)	(−0.37)	(−1.04)	(0.02)
Kink at SUE = 0, far-below firms	0.95**	0.86	0.92*	1.12	−0.23	1.01
	(2.04)	(0.98)	(1.73)	(0.85)	(−1.58)	(1.18)
Extra kink, at-record firms	−0.25	5.05***	1.94**	4.84**	1.08***	3.84**
	(−0.21)	(2.92)	(2.16)	(2.31)	(6.26)	(2.15)
Bandwidth h	1	1	2	1	1	1
Far-below firms: previous SUE > 0				yes		
Controls for next-quarter SUE						yes
Reports	50,620	48,991	53,624	26,370	51,247	48,819
at-record firms	10,203	10,000	14,655	10,000	10,263	9,967

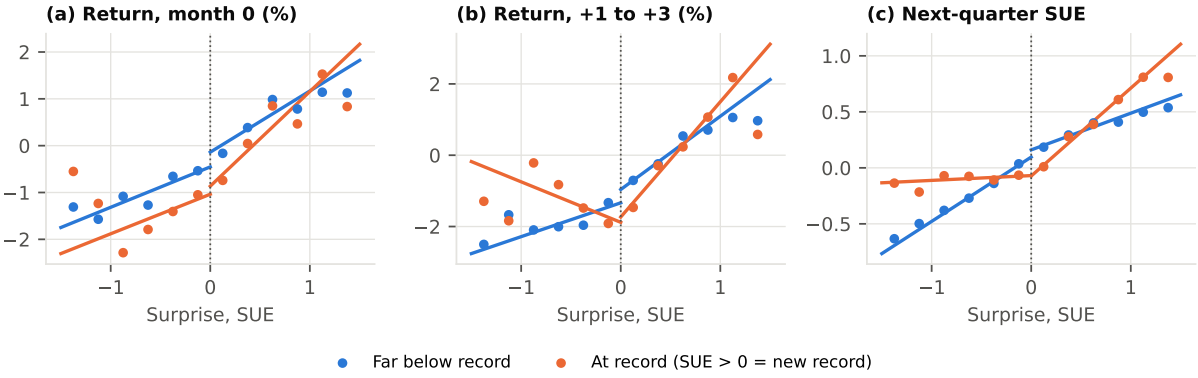


Figure 6: Response to the surprise by record status. Bin means net of report-month effects and local-linear fits on each side of $SUE = 0$. For firms at their record, $SUE > 0$ sets a new record.

Equation (4): at-record firms ($gap = 0$) versus firms far below their record ($gap > h$); report-month fixed effects; triangular kernel. Rows "Extra" are the differences between the two groups. Returns in % per unit of σ (slopes and kinks) or % (jumps); column 5 in units of the next-quarter SUE. t-statistics clustered by report month and firm.

delisting, and firms manage earnings to avoid them (Chen and Yuan, 2004; Haw et al., 2005). Firms also bunch just above $SUE = 0$, avoiding a decline from last year's quarter (Burgstahler and Dichev, 1997). They do so whether or not a record is at stake: firms at their record show 43% excess mass, and firms below their record that also had good news in the previous quarter show 32%.

The density test is less precise near the record than near zero earnings, because the density of r peaks close to the record. At placebo cutoffs between −2 and 2 the test finds "effects" with $|t|$ up to 3.20. The estimate at the true record is well inside this range, and in all specifications it is small compared with the bunching at zero earnings.

5.5 Robustness and heterogeneity

Figure 8 and Appendix Table A1 repeat the main design in 24 variations. These include different bandwidths, kernels and surprise bins, surprise-bin-by-year effects, surprises measured with an

Table 7: Bunching tests

	Linear, ±1		Quadratic, ±1		Linear, ±2	
	Excess (%)	Reports	Excess (%)	Reports	Excess (%)	Reports
Record, all reports	3.6 (1.11)	23,096	3.5 (0.70)	23,096	4.1 (1.56)	39,814
Record, firms with gap ≥ 1 (window ±0.5)	5.7 (0.45)	1,699	−4.8 (−0.31)	1,699	8.2 (0.68)	1,699
Beat last year (SUE = 0), firms at their record	42.8*** (4.83)	10,212	43.5*** (2.90)	10,212	51.0*** (7.78)	15,027
Beat last year (SUE = 0), firms below, previous SUE > 0	31.9*** (3.89)	21,323	22.8*** (5.44)	21,323	35.0*** (3.23)	26,667
Zero earnings (TTM EPS = 0), all reports	153.3*** (11.04)	17,802	104.9*** (8.27)	17,802	172.1*** (13.67)	29,117
Zero annual earnings, annual reports	1394.4*** (26.05)	4,535	1231.5*** (16.28)	4,535	1114.6*** (32.86)	7,076

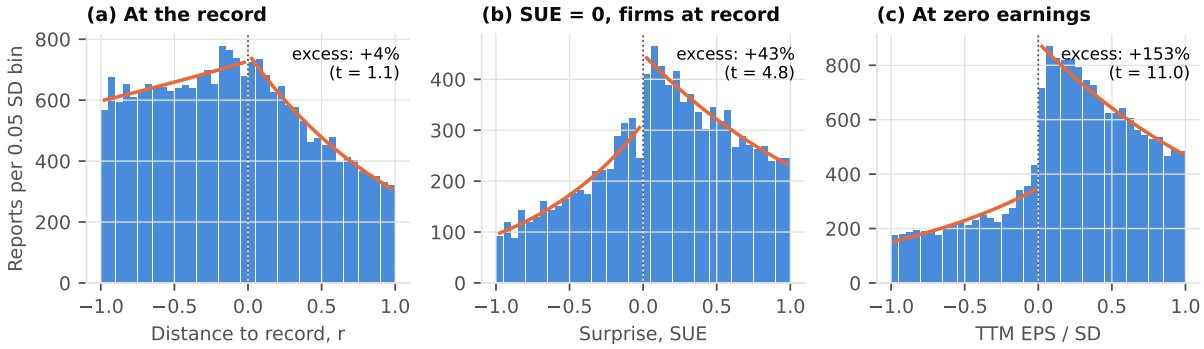


Figure 7: Reports per bin around three thresholds, with fitted densities on each side. (a) The record, all reports. (b) Beating last year's quarter, firms at their record. (c) Zero TTM earnings.
Poisson regressions of counts in bins of 0.05σ (0.025σ in row 2) on a polynomial on each side of the threshold, within the stated window. Excess mass $= e^{\hat{\delta}} - 1$ in %, with t-statistics (robust) in parentheses.

out-of-sample XGBoost forecast (Chen and Guestrin, 2016) instead of the SRW, a donut that drops reports within 0.05σ of the record, and controls. Attention constraints should matter most for small firms and firms without analyst coverage, and a record should be most salient when it is regained after a long time. None of these subsamples shows a robust effect. The results also hold for three other definitions of a record: the best single quarter, the best year-to-date result for the same period of earlier years (Chinese firms report cumulative figures) and the best annual result. Across 72 estimates, 3 are significant at the 5% level, about the 3.6 that chance alone would produce. The point estimates for the three-month drift are mostly positive, up to 2.4%, but significant in only two of the 24 rows. Placebo cutoffs on either side of the record give estimates of the same size as at the true record.

As a benchmark we apply the same design to the zero-earnings threshold, where Table 7 shows that firms care a great deal. Here too the announcement return (0.86%, $t = 1.73$) and turnover (7.88%, $t = 1.97$) jump only marginally. Monthly data can detect only large threshold effects, a point we return to below.

6 Discussion

Why does the record not matter, when price anchors such as the 52-week high do? Three differences seem important. First, a price anchor is visible every day on every trading screen, while an earnings record must be computed from past reports and is reported only when it is broken. Investors see a record headline only after the fact, together with the news that produced it, and the news is what the price responds to. Second, Chinese reports give cumulative year-to-date figures and growth on the same period of last year. Trailing records may simply not be the numbers investors look at. Our results for year-to-date and annual records suggest that this is not the whole story. Third, the record carries no information about future earnings beyond the surprise (Table 4), so there is nothing for an informed investor to trade on, and nothing for managers to

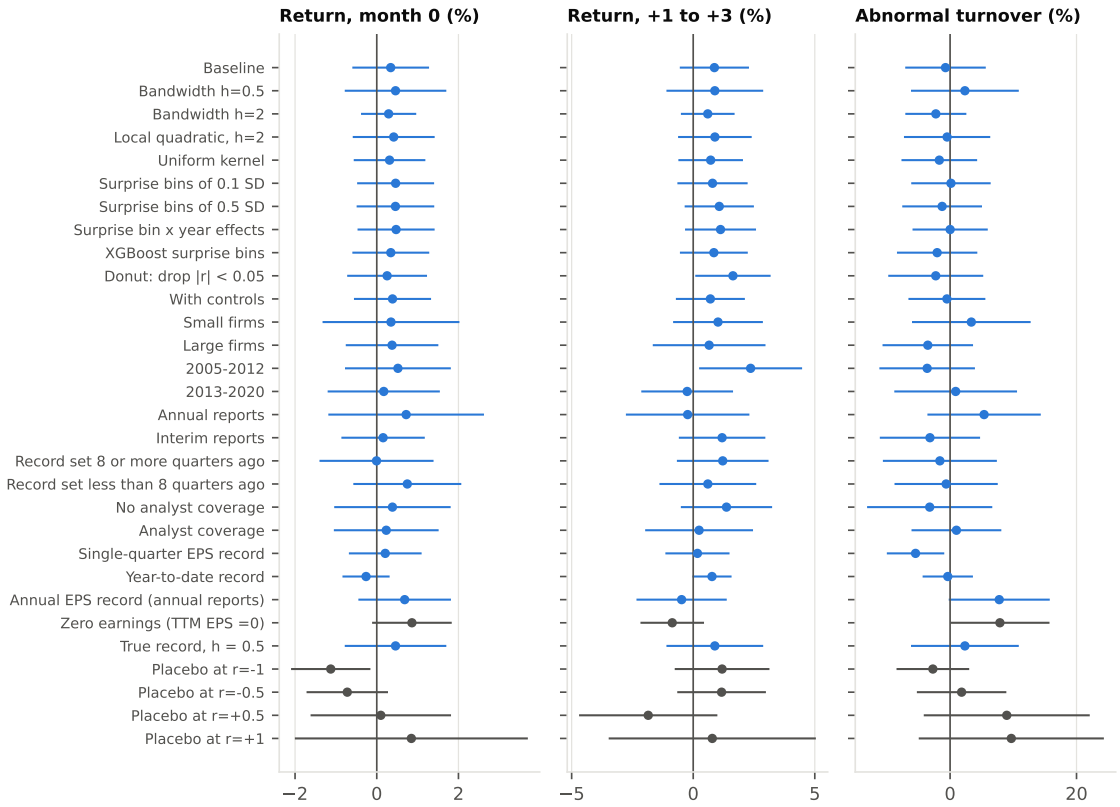


Figure 8: Robustness. Jump at the record from Equation (3) with 95% confidence intervals, for the specifications and samples of Appendix Table A1. Grey: benchmark threshold and placebo cutoffs.

gain from reaching it.

Our evidence has limits. Monthly returns cannot detect a premium that appears and disappears within a few days, and the benchmark at zero earnings shows that our design detects only large threshold effects in prices and volume. The bound we report is therefore a bound on effects that last at least until the end of the report month. Records are defined on split-adjusted EPS. Headlines usually refer to net profit, which differs from EPS when firms issue shares, so some firms are misclassified close to the record; such errors would weaken a true jump. Finally, the comparison of firms at their record with firms far below it is between two kinds of firms. We use it to describe how their news differs, not to identify a causal effect of the record.

7 Conclusion

Record earnings look like an ideal reference point, and a simple RD seems to confirm that investors anchor on them. The confirmation is an artefact. Because trailing earnings move by exactly the surprise, the distance to the record is the surprise minus the pre-report gap, and most records are set by firms that were already at their record. Using the gap, which is fixed before the report, to compare firms with the same news on either side of their record, we find that a new record moves neither prices, nor trading, nor analysts, nor future earnings, and that managers do not manage earnings to reach it. What does differ at the record is the persistence of news, which prices learn slowly. The method carries over to any threshold defined by a running maximum: before reading a break at the maximum as behaviour, ask whether the maximum was crossed by the news itself.

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Appendix

Table A1: Robustness, heterogeneity, other record definitions and placebos

	Return, month 0 (%)	Return, +1 to +3 (%)	Abnormal turnover (%)
Specification			
Baseline ($h = 1$, 0.25 SD surprise bins, triangular kernel)	0.34 (0.71)	0.87 (1.20)	−0.73 (−0.22)
Bandwidth $h = 0.5$	0.46 (0.72)	0.89 (0.88)	2.34 (0.54)
Bandwidth $h = 2$	0.29 (0.84)	0.60 (1.06)	−2.26 (−0.92)
Local quadratic, $h = 2$	0.41 (0.81)	0.89 (1.15)	−0.47 (−0.14)
Uniform kernel	0.31 (0.70)	0.72 (1.06)	−1.71 (−0.56)
Surprise bins of 0.1 SD	0.46 (0.96)	0.79 (1.08)	0.14 (0.04)
Surprise bins of 0.5 SD	0.46 (0.94)	1.07 (1.48)	−1.26 (−0.39)
Surprise bin × year effects	0.47 (0.98)	1.12 (1.51)	0.01 (0.00)
XGBoost surprise bins	0.34 (0.72)	0.85 (1.19)	−2.05 (−0.63)
Donut: drop $\|r\| < 0.05$	0.25 (0.51)	1.63** (2.07)	−2.27 (−0.59)
Controls: previous SUE, size, momentum, price/52w high	0.39 (0.80)	0.71 (0.97)	−0.51 (−0.16)
Subsamples			
Small firms	0.35 (0.41)	1.02 (1.08)	3.36 (0.70)
Large firms	0.37 (0.65)	0.65 (0.55)	−3.53 (−0.97)
2005–2012	0.52 (0.78)	2.36** (2.19)	−3.63 (−0.94)
2013–2020	0.17 (0.24)	−0.25 (−0.26)	0.88 (0.18)
Annual reports	0.72 (0.74)	−0.23 (−0.18)	5.38 (1.18)
Interim reports	0.16 (0.30)	1.19 (1.31)	−3.19 (−0.79)
Record set 8 or more quarters ago	−0.01 (−0.01)	1.21 (1.26)	−1.62 (−0.35)
Record set less than 8 quarters ago	0.75 (1.11)	0.60 (0.59)	−0.63 (−0.15)
No analyst coverage	0.38 (0.53)	1.37 (1.43)	−3.23 (−0.64)
Analyst coverage	0.23 (0.36)	0.24 (0.21)	1.00 (0.28)
Other record definitions			
Single-quarter EPS record	0.21 (0.46)	0.17 (0.26)	−5.47** (−2.36)
Year-to-date EPS record (same period of earlier years)	−0.26 (−0.89)	0.77* (1.88)	−0.37 (−0.18)
Annual EPS record (annual reports)	0.68 (1.18)	−0.48 (−0.50)	7.77* (1.91)
Benchmark threshold			
Zero earnings (TTM EPS = 0)	0.86* (1.73)	−0.86 (−1.29)	7.88** (1.97)
Placebo cutoffs (one side of the record only, $h = 0.5$)			
Baseline at the true record, $h = 0.5$	0.46 (0.72)	0.89 (0.88)	2.34 (0.54)
Placebo at $r = -1$	−1.13** (−2.27)	1.19 (1.19)	−2.73 (−0.93)
Placebo at $r = -0.5$	−0.72 (−1.42)	1.17 (1.25)	1.82 (0.50)
Placebo at $r = +0.5$	0.10 (0.11)	−1.85 (−1.28)	8.96 (1.34)
Placebo at $r = +1$	0.85 (0.58)	0.78 (0.36)	9.68 (1.30)

Each row re-estimates the jump at the record of Equation (3) (firms below their record, $h = 1$, surprise bins of 0.25σ, triangular kernel) unless stated otherwise; t-statistics in parentheses, clustered by report month and firm. Small and large firms split at the yearly median of size. Other record definitions use all reports with that definition and drop firms exactly at their record. The zero-earnings row uses TTM EPS/σ as the running variable. Placebo rows use reports on one side of the record only, with a bandwidth of 0.5.